

CHAYKIDAR: COMPLETE IOT SECURITY SYSTEM

Next-generation protection for smart homes and connected devices



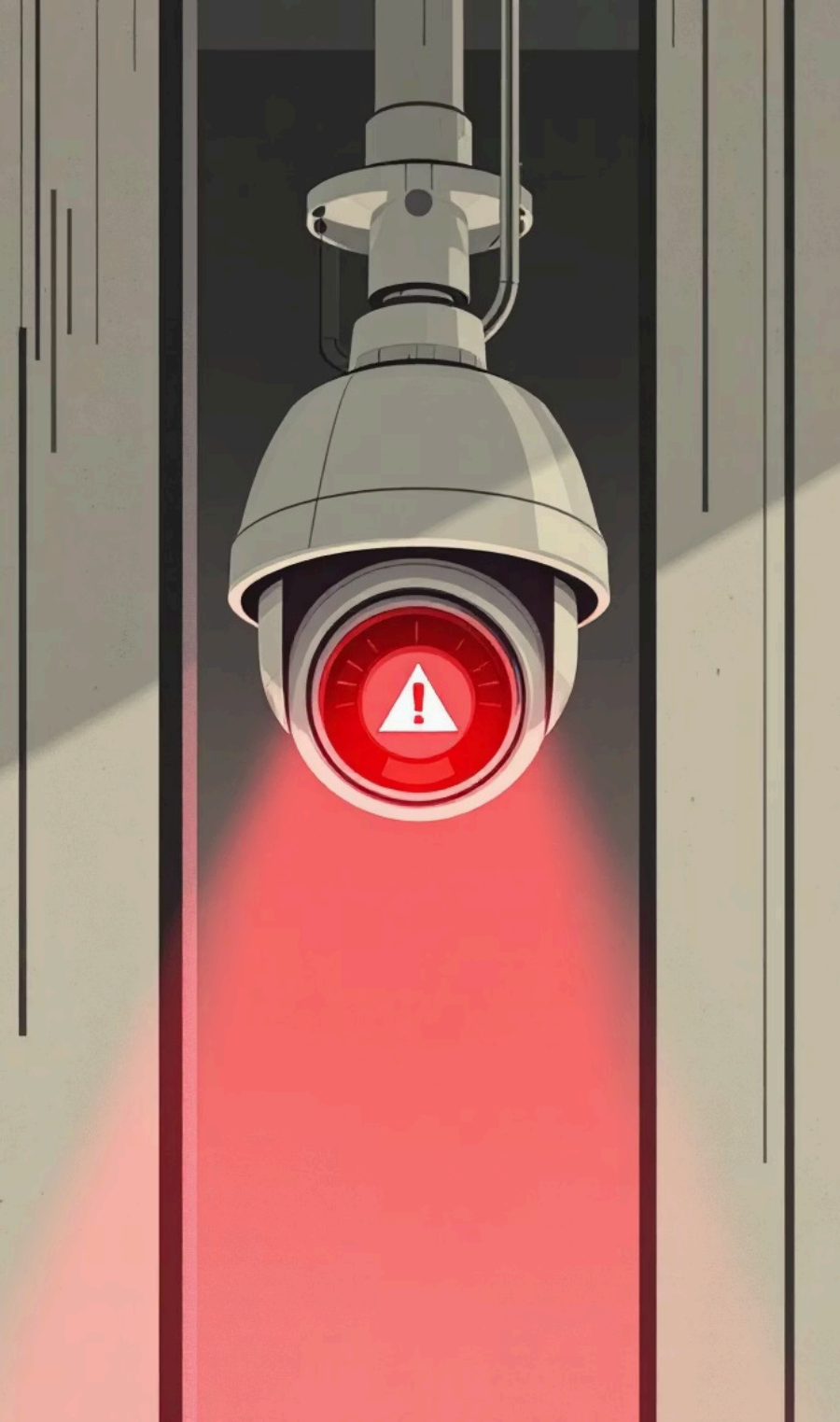
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SECURING THE INTERNET OF THINGS

CHAYKIDAR

Autonomous IoT security platform delivering next-generation protection for smart homes and connected devices.





CRITICAL THREAT

THE IOT SECURITY CRISIS

60B

**IOT DEVICES
BY 2025**

Most remain vulnerable to
attacks

72%

**ATTACKS FROM
INSIDE**

Originate within the
network

10+

**VULNERABLE
DEVICES**

Average per smart home

TOP ATTACK VECTORS

- Unpatched vulnerabilities (31%)
- Physical access exploits (27%)
- Default credentials (23%)
- Weak protocols (19%)

REAL-WORLD IMPACTS

- Camera hacks and privacy violations
- Smart lock bypasses
- Botnet recruitment (Mirai)
- Data exfiltration and ransomware

INTRODUCING CHAYKIDAR

THE GUARDIAN OF YOUR SMART HOME



Sanskrit for "Guardian" or "Watchman"

AUTONOMOUS SECURITY

Platform for IoT networks with zero-configuration protection

AI-DRIVEN DETECTION

Intelligent threat detection and automated response

"ASSUME BREACH" MINDSET

Always prepared for threats

DEFENCE-IN-DEPTH

Multi-layered architecture

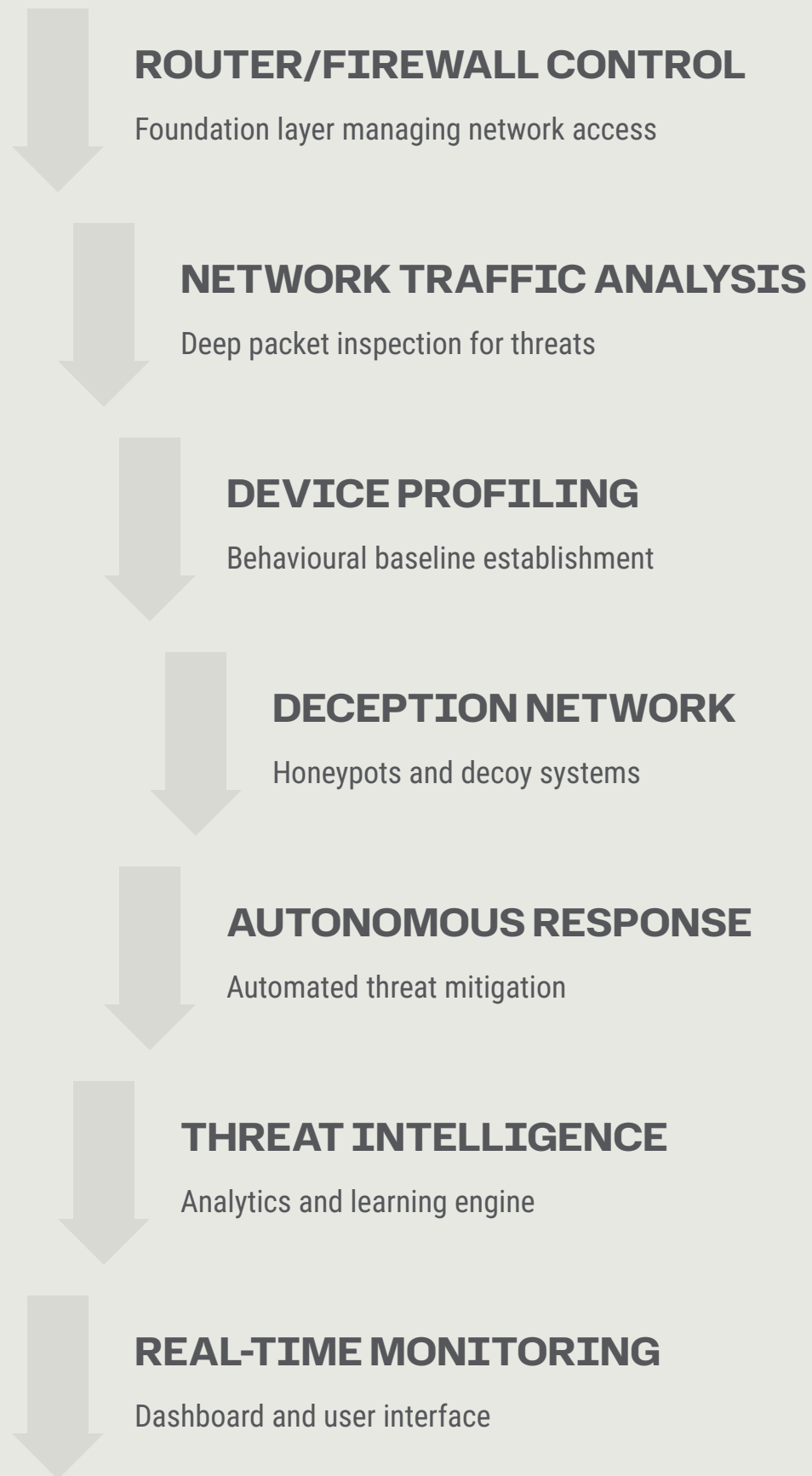
AUTONOMOUS RESPONSE

Immediate threat mitigation

DECEPTION-BASED SECURITY

Active attacker misdirection

MULTI-LAYERED DEFENCE SYSTEM



KEY FEATURES & CAPABILITIES

AUTOMATED DEVICE DISCOVERY

Network-wide scanning with device fingerprinting, vulnerability matching, and behavioural baseline establishment.

REAL-TIME THREAT DETECTION

ML-powered anomaly detection, protocol exploit identification, and reconnaissance monitoring.

ACTIVE DECEPTION DEFENCE

Honeypot IoT devices, honeytokens, decoy networks, and attack intelligence gathering.

⚡ RESPONSE TIME: < 2 SECONDS

AUTONOMOUS PROTECTION ENGINE

Automatic threat response system with intelligent detection and immediate action.

Threat Type	Detection Method	Response Action
Port Scanning	Traffic pattern analysis	Block source IP, isolate device
Brute Force	Failed auth monitoring	Rate limit, temporary block
IoT Exploit	Protocol anomaly detection	Patch virtual, alert admin
Data Exfiltration	Behavioural deviation	Block traffic, notify
Malware C2	DNS request monitoring	Sinkhole domain

ROUTER INTEGRATION METHODS

MULTIPLE CONTROL PATHWAYS



API INTEGRATION

Modern routers with direct REST API access, real-time rule updates, and full visibility.



SSH AUTOMATION

OpenWRT/DD-WRT script-based management with cron jobs and configuration backup.



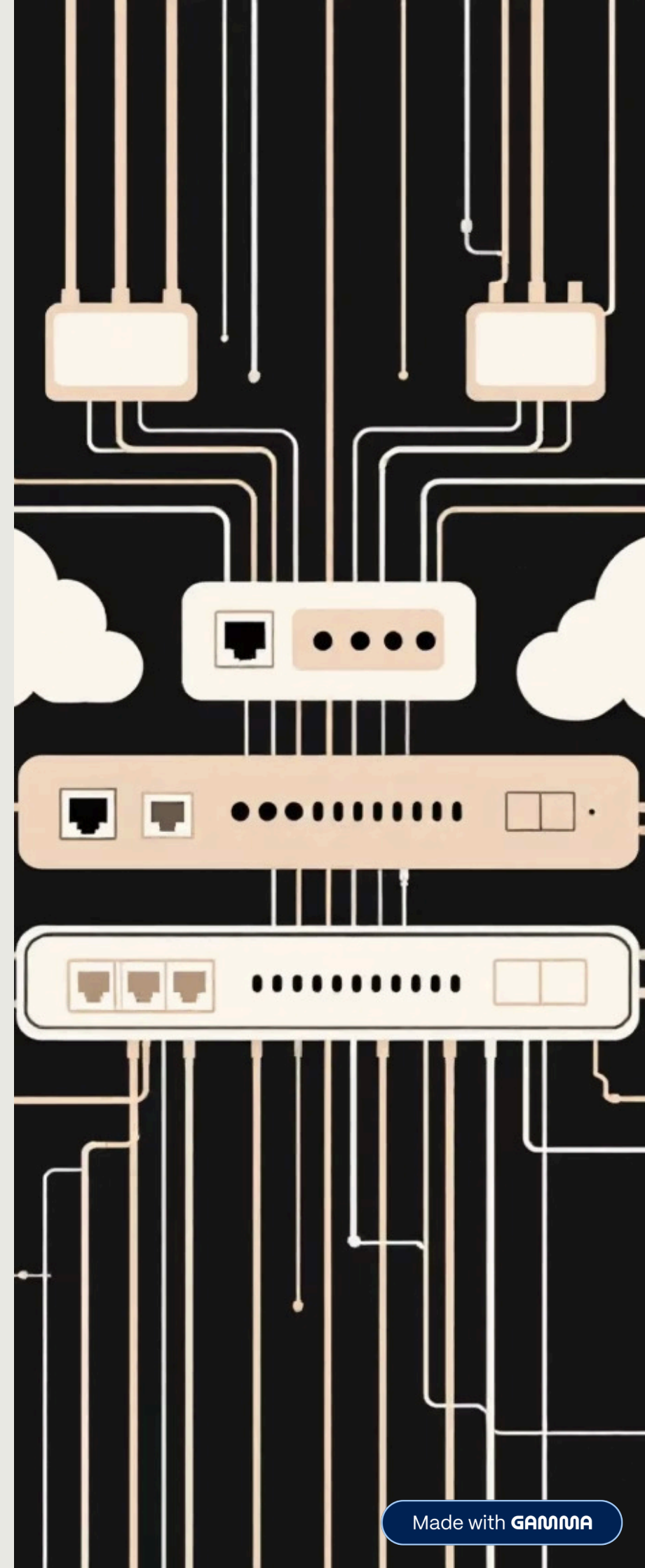
ARP-BASED CONTROL

Universal compatibility using ARP spoofing for traffic redirection and transparent protection.



DNS FILTERING

Local DNS server with malicious domain blocking and C2 traffic sinkholing.





HARDWARE DEPLOYMENT OPTIONS

CHOOSE YOUR DEPLOYMENT



RASPBERRY PI APPLIANCE

- Cost: £50-£100
- Power consumption: 5W
- Installation: Plug and play
- Features: Full protection suite



VIRTUAL MACHINE

- Platform: Proxmox, ESXi, Hyper-V
- Resources: 2GB RAM, 20GB storage
- Management: Remote access



CLOUD SUBSCRIPTION

- Model: SaaS monthly subscription
- Features: Remote monitoring
- Support: 24/7 threat intel updates

TECHNICAL IMPLEMENTATION

BACKEND STACK

- **DISCOVERY**

ARP + MAC + Port Scanning

- **ANALYSIS**

Deep Packet Inspection

- **DETECTION**

Machine Learning Models

- **RESPONSE**

Automated iptables/arp rules

- **DECEPTION**

Containerised honeypots

Technologies: Node.js, Express, Socket.io, Python, Docker, SQLite

FRONTEND STACK



REACT.JS

Modern UI framework



MATERIAL-UI

Consistent design system



CHART.JS

Real-time visualisations



WEBSOCKET

Live updates